Memory Dynamics and Higher-Order Observer Loops as Structural Conditions for Human-Like Intelligence: An Active Inference Perspective on Consciousness, Intelligence, and Personality

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Abstract

The free energy principle and active inference provide an influential framework for understanding adaptive living systems as self-organizing processes that maintain viability through perception, action, and model updating. However, this common formal substrate does not by itself explain the differences between plant-like regulation, animal cognition, and human abstract intelligence. This paper proposes that human-like intelligence requires two additional structural conditions built upon active inference: temporally layered memory dynamics and higher-order observer loops. Memory dynamics determine which representations decay, reactivate, consolidate, and become abstracted across time. Higher-order observer loops select, compare, maintain, and regulate these representations for flexible action, self-monitoring, and symbolic communication. Within this framework, intelligence is defined as the capacity of a resource-limited adaptive system to preserve, recombine, and deploy structural invariants for viable action; consciousness is defined as a dynamic mode in which selected representations become stabilized, self-referenced, and available for flexible control; and personality is defined as the relatively stable configuration of priors, precision-weighting policies, affective dispositions, memory-accessibility patterns, and self-narratives that shape how an agent interprets uncertainty over time. The paper argues that abstraction is not an isolated cognitive faculty, but an emergent consequence of lossy memory, selective consolidation, predictive control, and social-symbolic transmission. The framework is intended as an integrative hypothesis rather than a completed theory, and it generates testable predictions about memory consolidation, metacognitive control, personality stability, and artificial intelligence systems.

Keywords: active inference; free energy principle; memory dynamics; consciousness; personality; abstraction; predictive processing; human-like intelligence; higher-order control

1 Introduction

Contemporary cognitive science contains several powerful but partially disconnected accounts of intelligence, consciousness, memory, and personality. Predictive processing and active inference describe perception and action as processes of prediction-error minimization and uncertainty reduction (Clark, 2013; Friston, 2010; Friston et al., 2017). Memory research shows that remembering is not passive storage but selective, reconstructive, and shaped by consolidation (Conway & Pleydell-Pearce, 2000; McClelland et al., 1995; Nadel & Moscovitch, 1997). Theories of consciousness emphasize global availability, recurrent processing, higher-order representation, integrated information, or predictive self-modeling (Baars, 1988; Dehaene & Changeux, 2011; Lamme, 2006; Lau & Rosenthal, 2011; Tononi, 2004). Personality psychology describes relatively stable patterns of affect, cognition, motivation, and behavior (McCrae & Costa, 2008; DeYoung, 2015).

Yet these domains are often treated as separate explanatory projects. Intelligence is discussed in terms of learning and problem solving; consciousness in terms of subjective experience or access; personality in terms of traits; and memory in terms of encoding, storage, and retrieval. This paper proposes that these phenomena can be partially unified by viewing them as different time-scale expressions of a resource-limited adaptive system.

The argument can be summarized as a three-layer architecture. First, active inference provides the adaptive substrate: organisms must regulate themselves under uncertainty. Second, memory dynamics provide temporal selection: traces decay, reactivate, reconstruct, and consolidate, causing some structures to persist while others disappear. Third, higher-order observer loops and symbolic communication provide reflective control and social scaling: selected representations can be monitored, compared, self-referenced, and transmitted. The paper argues that intelligence, consciousness, and personality are different time-scale expressions of this layered architecture.

The central proposal is that human-like intelligence is not produced by prediction alone. It requires temporally layered memory dynamics and higher-order observer loops that determine which representations are forgotten, stabilized, abstracted, self-referenced, and socially transmitted.

The free energy principle provides the broadest background condition: adaptive systems must maintain their organization under uncertainty. However, this alone does not distinguish a plant, an insect, a mammal, a human being, or a language model. The relevant differences arise from structural additions: action channels, centralized control, offline simulation, abstract memory, higher-order monitoring, and symbolic externalization.

This paper develops a framework in which active inference provides the common adaptive substrate; memory dynamics explain the selective survival of useful representations across time; higher-order observer loops explain metacognitive control, self-monitoring, and abstraction; symbolic communication explains how individual abstractions become socially transmissible

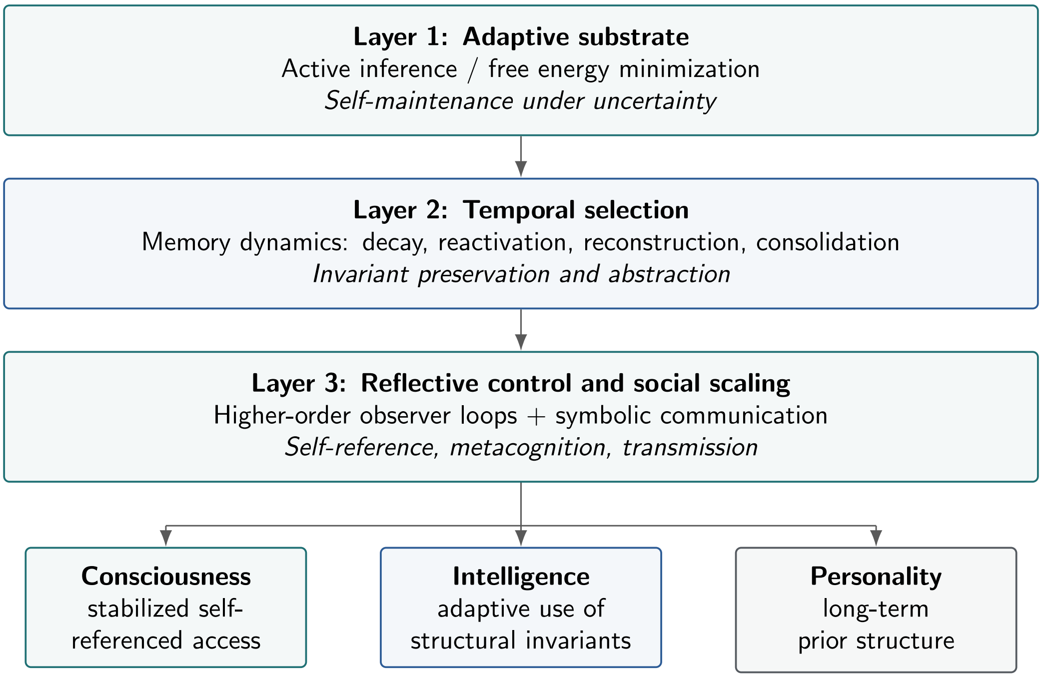

Figure 1: A three-layer architecture for human-like intelligence. Active inference provides the adaptive substrate, memory dynamics provide temporal selection and abstraction, and higher-order observer loops with symbolic communication provide reflective control and social scaling. Consciousness, intelligence, and personality are interpreted as different time-scale expressions of this layered architecture.

knowledge; and consciousness, intelligence, and personality are interpreted as different modes or time-scales of this same architecture.

The goal is not to reduce consciousness or personality to a single mechanism, nor to claim that all adaptive systems are conscious. Rather, the goal is to identify structural conditions under which resource-limited systems become capable of human-like abstraction, self-monitoring, and stable individuality.

2 Relation to Existing Frameworks

The present framework is not intended to replace existing theories of active inference, consciousness, memory, or personality. Rather, it attempts to specify a structural bridge among them. Compared with standard active inference accounts, the framework emphasizes memory decay, reactivation, and consolidation as mechanisms of abstraction (Friston, 2010; Friston et al., 2017). Compared with global workspace and higher-order theories of consciousness, it emphasizes the historical shaping of conscious contents by memory dynamics (Baars, 1988; Dehaene & Changeux, 2011; Lau & Rosenthal, 2011). Compared with trait theories of personality, it interprets traits as relatively stable priors and precision-weighting policies (McCrae & Costa, 2008; DeYoung, 2015). Compared with complementary learning systems theory, it extends the fast-slow memory distinction toward metacognitive observer loops and symbolic transmission

(McClelland et al., 1995; Kumaran et al., 2016).

3 Active Inference as a Common Adaptive Substrate

The free energy principle proposes that self-organizing systems maintain their integrity by minimizing a bound on surprise, or more precisely by acting to remain within expected states compatible with their continued existence (Friston, 2010). In active inference, perception updates beliefs to better predict sensory input, while action changes sensory input to better fit predicted or preferred states (Friston et al., 2017; Pezzulo et al., 2015).

At a coarse-grained level, this framework applies broadly to living systems. A cell maintains its boundary and internal chemical organization. A plant-like regulatory system can exploit environmental regularities through growth, chemical signaling, and distributed physiological adjustment. An animal moves, perceives, explores, avoids threat, and seeks resources. A human being does all of this while also constructing abstract models, social identities, symbolic institutions, and scientific theories.

The key point is that active inference describes a common logic of adaptive self-maintenance, but not the entire hierarchy of cognitive differences. A plant-like regulatory system and a human may both be describable as adaptive systems maintaining viability under uncertainty, but this does not mean that they possess the same cognitive architecture.

Therefore, the present framework distinguishes between a common substrate and structural increments. All living systems must maintain organization, regulate boundaries, and adapt to environmental uncertainty. Different forms of adaptive cognition may emerge when additional mechanisms are layered on top of this substrate.

These structural increments include morphological regulation, fast reversible action, centralized control, offline simulation, abstract memory, higher-order observer loops, and symbolic externalization. Morphological regulation supports slow adaptation through body form, growth, and distributed signaling. Fast reversible action allows an organism to sample and change its environment. Centralized control integrates sensory, motor, motivational, and regulatory signals. Offline simulation models absent or future states. Abstract memory preserves reusable relational structure rather than raw detail. Higher-order observer loops monitor, compare, select, and regulate other representational states. Symbolic externalization extends cognition through language, writing, diagrams, mathematics, institutions, and cultural memory.

This distinction avoids two opposite errors. The first error is to deny continuity between simple life and advanced cognition. The second is to collapse all adaptive regulation into the same category and thereby lose the specific structural conditions that make human cognition distinctive.

4 Memory Dynamics: Forgetting as Selective Survival

Memory is often described as the ability to store information. This description is incomplete. Biological memory is not a perfect archive. It is lossy, selective, reconstructive, emotionally weighted, and dependent on repeated reactivation and consolidation (Tulving, 1985; Baddeley, 2000; McClelland et al., 1995; Nadel & Moscovitch, 1997; Conway & Pleydell-Pearce, 2000; Kumaran et al., 2016).

The present framework treats memory as a dynamic selection process. In a resource-limited system, not all traces can be preserved equally. Energy, attention, synaptic stability, storage capacity, and computational access are limited. As a result, memory must solve a continual allocation problem: which traces should decay, which should remain accessible, which should be consolidated, and which should be reconstructed into more abstract forms?

A useful formal intuition is:

$$\dot{\theta} = -\lambda\theta + \alpha R_t - \eta \nabla_\theta \mathcal{L}_t + \gamma C_t, \tag{1}$$

where θ denotes a memory trace or model parameter, $-\lambda\theta$ denotes decay, R_t denotes reactivation, $\mathcal{L}_t$ denotes prediction or reconstruction loss, and C_t denotes consolidation pressure.

Equation 1 is not intended as a complete biological model. It is a conceptual compression of four broad memory forces: decay, reactivation, reconstruction, and consolidation. Decay means that unused traces lose accessibility or strength. Reactivation means that retrieved or rehearsed traces regain influence. Reconstruction means that traces are reshaped under current goals, predictions, and contexts. Consolidation means that repeatedly useful structures migrate into slower, more stable representational forms.

> **Illustrative note on formalization.** A simple leaky integrate-and-fire model provides a minimal example of how decay, input-driven activation, and thresholded event generation can be formalized in neural dynamics:
>
> $$\tau_m \frac{dV}{dt} = -(V - V_{\text{rest}}) + R_m I(t). \tag{2}$$
>
> Here, membrane potential V decays toward a resting state unless maintained by input $I(t)$, and crossing a threshold produces a discrete spike. This equation is not intended as a model of long-term memory consolidation. Rather, it illustrates a lower-level dynamical motif–leak, input-driven activation, and thresholded event generation–that can be generalized cautiously at higher levels to memory accessibility, representational persistence, and abstraction.

The implication is important: forgetting is not merely failure. It is part of the mechanism by which adaptive systems separate structure from noise.

A system that preserved every detail with equal fidelity would face severe computational and behavioral problems. It would be burdened by irrelevant particulars. By contrast, a lossy system that preferentially preserves repeated, predictive, low-cost, action-relevant patterns can gradually extract structural invariants.

Thus, abstraction is not an additional faculty placed on top of memory. It is a consequence of selective memory survival under resource constraints.

This does not mean that forgetting always produces truth. Memory dynamics can also preserve bias, trauma, superstition, or maladaptive priors. The claim is narrower: under repeated structured experience, adaptive feedback, and error correction, lossy memory tends to privilege reusable structure over accidental detail.

5 Abstraction as Invariant Preservation

Abstraction is usually described as removing irrelevant details from particular examples. The present framework reframes abstraction as the long-term result of differential persistence.

A representation becomes abstract when it survives across changing contexts because it captures something stable enough to be reused. For example, a child learning "dog" does not retain every visual detail of every encountered dog. Instead, through repeated exposure, error correction, and social labeling, the child gradually stabilizes a category structure. The category is not a copy of any single experience. It is a compressed invariant extracted across many experiences.

The same principle applies at higher levels. A physical concept such as force abstracts across many kinds of motion. A mathematical function abstracts across many concrete quantities. A moral rule abstracts across many social conflicts. A personality self-description abstracts across many episodes of behavior and feedback.

On this view, abstraction has three layers. Statistical abstraction detects repeated patterns across sensory or behavioral input. Predictive abstraction preserves patterns that reduce future prediction error or support useful action. Reflective abstraction compares, manipulates, and explicitly reasons about abstractions themselves. The ability to construct absent, possible, or future scenes is closely related to constructive episodic memory and simulation (Hassabis & Maguire, 2007).

The third layer is especially important for human cognition. Many animals can form categories and expectations. Humans, however, can take categories themselves as objects of thought. We can ask what a concept means, compare competing definitions, invent formal symbols, and modify our own reasoning strategies.

This reflective abstraction requires more than memory. It requires higher-order control.

6 Higher-Order Observer Loops

The phrase higher-order observer loop does not refer to a single organ that literally watches the entire brain. It refers to a functional control architecture.

Definition 1. *A higher-order observer loop is a control process that selects, compares, maintains, reactivates, and regulates representational states in relation to uncertainty, goals, action, and the self-model.*

Such a loop can select representations for further processing, compare competing representations, maintain information across time, reactivate relevant memories, integrate multimodal or cross-context information, detect conflict or uncertainty, regulate action and attention, and construct self-referential interpretations.

In the human brain, these functions likely depend on distributed networks involving prefrontal cortex, anterior cingulate cortex, parietal regions, hippocampal systems, salience networks, and other control systems. The prefrontal cortex is not a container of abstract knowledge by itself. Rather, it participates in maintaining goals, rules, task states, relational structures, and long-range control (Botvinick et al., 2001; O'Reilly & Frank, 2006).

This distinction matters. The framework does not claim that the prefrontal cortex is the observer. Instead, it claims that human-like abstraction and metacognition require control loops capable of selecting and regulating memory-based representations.

In simplified form:

$$O_t = f(M_t, E_t, G_t, S_t), \tag{3}$$

where O_t is the observer/control state, M_t is memory state, E_t is prediction error or uncertainty, G_t is goal or preference structure, and S_t is self-model state.

The observer loop then modulates:

$$(M_{t+1}, A_{t+1}, P_{t+1}, S_{t+1}), \tag{4}$$

where M is memory, A is action, P is precision weighting or attention allocation, and S is the self-model.

This architecture changes the nature of learning. A system without higher-order control can still learn associations. But a system with higher-order observer loops can decide what to inspect, what to rehearse, what to inhibit, what to compare, and what kind of error matters. This is the transition from passive learning to reflective learning.

7 Consciousness as Stabilized, Self-Referenced Access

Theories of consciousness remain highly contested. Global workspace theories emphasize broad access and broadcasting (Baars, 1988; Dehaene & Changeux, 2011). Recurrent processing theories emphasize feedback loops (Lamme, 2006). Higher-order theories emphasize representations about representations (Lau & Rosenthal, 2011). Integrated information theory emphasizes intrinsic causal integration (Tononi, 2004). Predictive processing approaches emphasize self-modeling, prediction, and embodied control (Clark, 2013; Seth, 2013; Seth & Friston, 2016).

The present framework does not attempt to replace these theories. Instead, it proposes a bridge: consciousness is a dynamic mode in which selected representations become stabilized, self-referenced, and available for flexible control.

This functional framing is compatible with approaches that treat consciousness as a distributed, interpretive, and access-dependent process rather than a single inner theater (Dennett, 1991).

This definition has four components. Selection means that not all neural activity becomes conscious. Conscious contents are selected from many competing internal and external signals. Stabilization means that conscious contents must persist long enough to be integrated, reported, remembered, or used. Self-reference means that conscious contents are situated relative to a self-model: this is happening to me, I am seeing this, I am deciding this, or I am uncertain about this. Flexible control means that conscious contents can guide deliberate action, reasoning, planning, communication, or metacognitive evaluation.

This account is compatible with several mainstream theories. From the perspective of global workspace theory, consciousness requires availability to multiple cognitive systems. From the perspective of higher-order theory, consciousness requires some form of meta-representation. From the perspective of active inference, consciousness is tied to self-modeling and precision-weighted control. From the perspective of recurrent processing, consciousness requires feedback stabilization.

The proposed contribution is to link these mechanisms to memory dynamics. Consciousness is not only a matter of immediate access. It also depends on which representations have been shaped by previous decay, consolidation, reactivation, and abstraction.

A conscious thought is therefore not simply a current state. It is a current stabilization of a history-shaped representational system.

8 Intelligence as Adaptive Use of Structural Invariants

In this framework, intelligence is not defined as raw computation, problem solving, or linguistic fluency alone. Intelligence is the capacity of a resource-limited adaptive system to selectively preserve, recombine, and deploy structural invariants in order to reduce future uncertainty and support viable action.

This definition has several advantages. First, it includes biological intelligence without reducing intelligence to human reasoning. A plant-like regulatory system can exploit environmental regularities through growth, chemical signaling, and distributed physiological adjustment. An animal can learn sensorimotor contingencies. A human can construct abstract theories. Second, it emphasizes resource limits. Intelligence is not perfect representation. It is adaptive compression. Third, it connects memory and action. A system is intelligent not because it stores the past, but because it uses selected structure from the past to act under future uncertainty.

The definition also distinguishes between different intelligence types. Regulatory intelligence maintains internal viability through physiological or distributed control. Sensorimotor intelligence uses perception and action to navigate changing environments. Simulative intelligence uses internal models to evaluate absent, counterfactual, or future states. Abstract intelligence uses compressed relational structures across contexts. Reflective intelligence monitors and modifies one's own models, memories, strategies, and beliefs. The term intelligence is used here in a broad adaptive sense, not as a claim that all regulatory systems possess consciousness or human-like cognition.

Human-like intelligence requires the last two layers to become strongly developed and socially externalized.

9 Personality as Long-Term Prior Structure

Personality can be defined as a relatively stable pattern of affect, cognition, motivation, and behavior. Trait models such as the Big Five describe these patterns statistically (McCrae & Costa, 2008). The present framework offers a process-level interpretation: personality is the relatively stable configuration of priors, precision-weighting policies, affective dispositions, memory-accessibility patterns, and self-narratives that shape how an agent interprets uncertainty and selects action over time (DeYoung, 2015).

This definition translates personality into active inference language. A highly anxious personality style may involve high threat priors, strong precision assigned to negative cues, rapid retrieval of danger-related memories, low tolerance for uncertainty, strong prediction-error responses to ambiguous social signals, and self-narratives organized around vulnerability or anticipated loss.

A highly open personality style may involve higher expected value of exploration, lower threat weighting for novelty, flexible category boundaries, tolerance for model revision, and greater reward from pattern discovery and cognitive recombination.

A highly conscientious personality style may involve strong goal maintenance, stable action policies, high precision assigned to future consequences, stronger self-regulatory observer loops, and memory structures organized around obligations, plans, and standards.

This does not replace trait psychology. Rather, it gives traits a mechanistic interpretation. Traits are not merely descriptive factors; they may reflect stable configurations of predictive priors, affective valuation, memory accessibility, and control policy.

Personality also depends on time scale. Short-term moods fluctuate rapidly. Habits stabilize over weeks or months. Traits stabilize over years. Self-narratives may persist across decades. This makes personality a slow variable in the broader memory-observer system.

Thus, consciousness is the current-access mode of the system; personality is the long-term prior mode of the system; intelligence is the adaptive control capacity across modes.

10 Self as a Higher-Order Generative Model

The concept of self is often treated as mysterious because it appears to refer to an entity behind experience. The present framework instead treats self as a model: the self is a higher-order generative model that maintains bodily, behavioral, narrative, and social continuity across time (Conway & Pleydell-Pearce, 2000; McAdams, 2001; Seth, 2013).

This model includes several layers. The bodily self predicts and regulates interoceptive and proprioceptive states. The agentic self predicts one's own actions and their consequences. The emotional self interprets affective and motivational states. The narrative self organizes autobiographical memory into a coherent story. The social self represents how others perceive and respond to the agent. The reflective self evaluates one's own beliefs, emotions, and actions.

The self-model is not always accurate. It is useful because it provides continuity and control. It constrains memory reconstruction, stabilizes personality, organizes conscious experience, and supports social coordination.

This also explains why disruptions of self can affect consciousness, memory, and personality simultaneously. If the self-model changes, then the interpretation of memories, emotions, intentions, and social signals changes as well.

11 Language and the Social Scaling of Abstraction

Individual memory dynamics can produce abstraction, but human civilization requires more than individual abstraction. It requires externalization.

Language transforms internal abstractions into public symbols. Once a structure is encoded symbolically, it can be transmitted to another person, criticized, compared with alternatives, written down, taught, institutionalized, mathematically formalized, and preserved across generations.

This creates a second level of selection. At the individual level, memory dynamics determine which patterns survive within a mind. At the social level, communication dynamics determine which patterns survive across minds.

A concept that is useful, teachable, compressive, and socially coordinateable has a higher chance of cultural persistence. Scientific theories, legal rules, moral norms, mathematical systems, and religious narratives can all be understood as socially stabilized abstractions.

This does not mean that socially persistent ideas are always true. False beliefs can also persist if they are emotionally powerful, identity-supporting, politically useful, or easy to transmit. Therefore, civilization-level knowledge requires not only transmission but also correction mechanisms: debate, experiment, peer review, replication, education, and institutional memory.

Within this framework, civilizational knowledge is the cumulative result of individual memory selection, social transmission, symbolic compression, and institutional error correction.

12 Implications for Artificial Intelligence

Current large language models exhibit impressive abstract recombination in symbolic space. They can summarize, infer, translate, write code, produce analogies, and simulate many forms of reasoning. In the present framework, this indicates that they have acquired large-scale compressed statistical and semantic structure.

However, this does not by itself imply human-like consciousness, personality, or embodied intelligence. On the present framework, current large language models instantiate powerful forms of symbolic abstraction and recombination, but they lack several architectural conditions associated with human-like self-regulation, personality stability, and conscious access.

From the perspective of the present framework, the central limitation of current large language models is not simply the absence of an additional reasoning module, but the absence of an autonomous loop that selects, forgets, consolidates, and regulates memory in relation to self-maintenance, uncertainty, and action. Scaling parameters, training data, and context length may improve task performance, but they do not by themselves create the kind of autonomous memory-selection and self-regulatory architecture required by the present framework.

Current mainstream language models typically lack autonomous bodily regulation, intrinsic survival constraints, persistent self-maintenance, stable autobiographical memory, endogenous goals grounded in organismic needs, long-term self-model continuity, autonomous memory decay and consolidation, and higher-order observer loops with causal control over future learning.

A language model can simulate a personality, but this is not the same as possessing a personality in the psychological sense. A simulated personality is usually induced by training data, system prompts, reinforcement learning, and local context. A biological personality is a long-term configuration of priors, affective dispositions, memory accessibility, and self-narrative shaped by lived interaction.

Similarly, a model can report conscious-like states without necessarily possessing conscious experience. This distinction is especially important. The present framework does not claim that artificial systems cannot become conscious. It claims that human-like consciousness would require architectural conditions beyond next-token prediction alone.

A more human-like artificial system would require persistent memory with decay and consolidation, self-monitoring and uncertainty estimation, explicit conflict detection, value-sensitive memory updating, autonomous goal management, a self-model connected to action consequences, and higher-order control loops capable of regulating attention, retrieval, and learning.

This suggests a research direction: rather than merely scaling model size, AI systems may need memory dynamics and observer architectures that determine what is forgotten, what is retained, what is abstracted, and what is revised.

13 Testable Predictions

Although the present framework is theoretical, it generates several empirical and computational predictions.

Prediction 1: Abstraction should depend on selective forgetting. Learning conditions that encourage selective forgetting of irrelevant detail, while preserving predictive structure, should improve abstraction and transfer. Perfect retention of surface detail may impair generalization in some tasks. This could be tested using category-learning or rule-induction tasks in which surface details are manipulated independently from relational structure.

Prediction 2: Reactivation should shape abstraction. Repeated reactivation of structurally relevant memories should increase abstraction more than passive exposure alone, especially when reactivation occurs across varied contexts. This could be tested by comparing repeated retrieval, passive restudy, and cross-context rehearsal in abstraction-transfer tasks.

Prediction 3: Metacognitive control should modulate memory consolidation. Individuals with stronger metacognitive monitoring and executive control should show more adaptive consolidation of task-relevant structure and better suppression of irrelevant detail. This could be tested by combining metacognitive judgment tasks with delayed memory tests and measures of sleep-dependent consolidation.

Prediction 4: Personality traits should correspond to stable precision-weighting policies. Trait differences should be associated with stable differences in how individuals assign precision to threat cues, reward cues, social feedback, uncertainty, and self-relevant information. This could be tested by combining personality measures with computational modeling of precision weighting in threat, reward, and uncertainty-learning tasks.

Prediction 5: Disrupted self-models should affect both consciousness and personality. Conditions involving self-disturbance should show coupled changes in conscious access, autobiographical memory, affective regulation, and trait-like behavior. This could be tested in clinical or subclinical populations using measures of self-concept clarity, autobiographical memory coherence, metacognitive access, and affective regulation.

Prediction 6: Artificial agents with lossy memory and observer loops should generalize differently from standard language models. Agents equipped with selective decay, reactivation, consolidation, and metacognitive control should show improved long-term adaptation, calibration, and self-correction compared with systems that rely only on static pre-trained weights and short context windows. This could be tested computationally by comparing agents with static context windows against agents with explicit decay, replay, consolidation, and metacognitive retrieval control.

14 Limitations

This framework is intentionally integrative, and therefore faces several limitations.

First, it is not a complete neurobiological model. It does not specify all relevant circuits, neurotransmitter systems, oscillatory mechanisms, or developmental pathways.

Second, it does not solve the hard problem of consciousness. It offers a functional and mechanistic account of conscious access, self-reference, and control, but it does not claim to explain why subjective experience exists at the metaphysical level.

Third, it risks overgeneralization if applied too broadly. Not every adaptive system is conscious. Not every memory system produces useful abstraction. Not every socially persistent idea is knowledge.

Fourth, the concept of a higher-order observer loop must be operationalized carefully. Without clear measures, it could become metaphorical. Future work should define observer-loop strength in terms of conflict monitoring, memory reactivation control, precision modulation, self-model updating, and flexible behavioral regulation.

Fifth, the relationship between physical free energy, variational free energy, and psychological uncertainty must be handled with caution. These concepts are related within broader active inference traditions, but they are not interchangeable at every level of analysis.

Despite these limitations, the framework may be useful because it links several otherwise separate questions: how organisms adapt, how memories become abstractions, how consciousness becomes accessible, how personality stabilizes, and how symbolic culture accumulates knowledge.

15 Conclusion

The free energy principle and active inference describe a common adaptive logic of living systems: organisms must maintain themselves under uncertainty through perception, action, and model updating. But this common substrate does not by itself explain the distinctive features of human cognition.

This paper has proposed that human-like intelligence requires two additional structural conditions: temporally layered memory dynamics and higher-order observer loops. Memory dynamics determine which representations decay, reactivate, consolidate, and become abstracted. Higher-order observer loops determine which representations are selected, compared, self-referenced, maintained, and used for flexible control.

Within this framework, intelligence, consciousness, and personality are not separate substances or modules. They are different time-scale expressions of one adaptive architecture. Intelligence is the system's capacity to preserve, recombine, and deploy structural invariants for viable action. Consciousness is the dynamic mode in which selected representations become stabilized, self-referenced, and available for flexible control. Personality is the long-term configuration of priors, precision policies, affective dispositions, memory accessibility patterns, and self-narratives that shape how the system interprets uncertainty.

The deepest implication is that abstraction is not a luxury added after memory. It is what memory becomes when decay, reactivation, prediction, consolidation, and control operate under resource constraints. Human civilization then extends this process through language, writing, institutions, and cultural error correction.

Human-like intelligence, therefore, is not merely larger computation. It is organized forgetting, selective remembering, reflective control, and symbolic transmission across time.

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